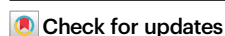


Interfacial W–O–Zr ensembles in tungstated zirconia catalysts enable efficient hydrogen-free recycling of polypropylene waste

Received: 30 December 2025

Accepted: 11 May 2026

Published online: 22 May 2026



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Polyolefins are essential plastics to modern life yet create mounting sustainability challenges. Chemical recycling technologies are key but often require harsh conditions, costly hydrogen, noble metals, or complex catalysts with challenging design. We report hydrogen- and solvent-free depolymerization of polypropylene consumer goods at 240 °C, achieving >80% yield into gasoline hydrocarbons with stable performance. A library of tungstated zirconia catalysts synthesized by flame spray pyrolysis with controlled tungsten speciation enables correlating performance with the density of W–O–Zr ensembles prevalent in sub-nanoclusters, with W:Zr = 1:9 emerging as optimal. These acid sites formed during reaction mediate internal hydrogen transfer and selective backbone C–C scission, as deduced from *operando* spectroscopy. Life cycle and technoeconomic studies, benchmarked against hydrotreatment-based recycling in a harmonized framework, indicate competitive environmental and economic advantages. Together, this work establishes this route's promise and underscores the value of precision catalyst synthesis in polyolefin valorization.

Polyolefins are plastics at the core of modern life with no foreseeable replacement in sight, making chemical recycling technologies play a central role in advancing a sustainable and circular society^{1–5}. Alternative strategies to the largely studied and commercially available pyrolysis^{6–8}, such as hydrotreatments have gained momentum in research laboratories worldwide due to their milder conditions and lighter downstream processing^{9–12}. Their practical deployment, however, relies on enhancing catalytic performance and resolving uncertainty related to future green hydrogen supply^{13–17}.

Hydrogen-free depolymerization at mild conditions has been recently demonstrated and so far focused on polyethylene¹⁸. Reports include alkane/polyolefin tandem cracking¹⁹, polyolefin dehydrocyclization and aromatization^{20–22}, and internal hydride transfer through Lewis acid site, LAS–Brønsted acid site, BAS synergy^{23,24}. For instance, aluminum chloride in ionic liquids with Lewis acidity enable tandem cleavage with isobutane or isopentane at temperatures as low

as 100 °C, yielding C₄–C₃₆ liquid alkanes¹⁹. Pt/Al₂O₃ promotes dehydrocyclization and aromatization offering incipient but encouraging performance, with halogenated supports accelerating tandem reactions by increasing acidity^{20–22}. More recently, WZr-KIT-6 (W- and Zr-incorporated KIT-6 silica materials) with HZSM-5 in physical mixtures were claimed to yielding C₄–C₁₂ hydrocarbons from polyethylene^{23,24}. In these systems, the WZr-KIT-6 component is thought to pre-crack polymer chains on LAS and BAS, while C–C bond scission and product selective shaping occurs predominantly within the micropores of HZSM-5. This study suggests the challenge of integrating efficient acid functions within a single catalytic surface²⁴.

These systems exhibit a degree of structural complexity hindering unambiguous structure-performance correlations. As an example, acidity is frequently invoked as a key catalytic factor^{21–24}; yet correlations between total acidity or simple Brønsted/Lewis ratios and performance have not been identified. Whether this reflects the

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complexity of the systems or a more fundamental aspect of the reaction chemistry remains unclear. The resulting uncertainty, compounded by a lack of *operando* studies, underscores the need for structurally well-defined catalysts that enable reliable structure-function assessment. A second critical knowledge gap lies in early feasibility evaluation, as few chemical recycling technologies for plastics have been benchmarked using harmonized environmental and economic metrics^{25–28}.

Here, we address these gaps by investigating the first efficient catalytic depolymerization of polypropylene (PP) in the absence of hydrogen or any other reductant or auxiliary means at mild temperatures^{29–35}. We develop a library of materials showing that zirconia-supported sub-nano clusters of tungsten generated via flame spray pyrolysis achieve over 90% polypropylene conversion with ~90% selectivity to gasoline-range hydrocarbons and identify W–O–Zr as active sites through hydrogen transfer^{36–38} via *operando* characterization. Process simulations coupled to sustainability assessments including hydrotreatments show that hydrogen-free PP chemical recycling offers environmental and economic advantages. Taken together, this work suggests interfacial M–O–M' (M = metal) linkages as a design principle for feasible pathways to polyolefin waste valorization.

Results

Catalyst synthesis strategy and polypropylene waste depolymerization

Flame spray pyrolysis, FSP was selected to synthesize a series of tungstated zirconia catalysts with well-defined structural features (denoted W_aZr_b , where *a* and *b* represent the mol% of tungsten and zirconium, respectively; Fig. 1a, Supplementary Table 1) in view of the versatility of this technique in producing materials with controlled architectures across wide compositional ranges^{39–41}. STEM-HAADF imaging exploiting the high *Z* contrast between W and Zr revealed that all samples featured ca. 10 nm ZrO_2 spheres (Supplementary Fig. 1), discarding support effects across the series, and decorated with distinct tungsten architectures. At low W contents (e.g., W_3Zr_{97}), uniformly dispersed isolated W atoms were prevalent with the presence of scattered sub-nanoclusters (Fig. 1b and Supplementary Figs. 2, 3). As the W content increased in $W_{10}Zr_{90}$, 0.3–0.8 nm WO_x sub-nanoclusters preferentially anchored at surface edge defects and grain boundaries of ZrO_2 emerged as the predominant speciation compared to isolated W centers (Fig. 1c and Supplementary Fig. 2)^{30,42}. At higher W contents, as in $W_{60}Zr_{40}$, irregular WO_3 nanoparticles (5–40 nm) populated the ZrO_2 surface accompanied by a minor fraction of sub-nanoclusters (Fig. 1d and Supplementary Fig. 3). X-ray diffraction, XRD analyses confirmed the gradual structural change characterizing the series (Supplementary Fig. 4).

Virgin PP with a weight average molecular weight, $M_w = 12$ kDa, denoted as PP_{12k}, was efficiently depolymerized over W_aZr_b catalysts under mild conditions similar to polyethylene reports (Fig. 2a)^{20–24}. Product distribution analysis revealed a volcano-type dependence between performance and tungsten content, with $W_{10}Zr_{90}$ giving >90% conversion and ca. 90% selectivity to gasoline range (C_4 – C_{12}) hydrocarbons (Fig. 2b, Supplementary Table 2, Supplementary Fig. 5)^{23,24}. Notably, experiments at different reaction times over this catalyst disclosed its high activity. After 10 min, 75% conversion with 88% gasoline selectivity was achieved^{43,44}, whereas prolonged durations of up to 24 h slowly shifted the product spectrum toward C_4 – C_5 light gases (Supplementary Figs. 6, 7 and Supplementary Table 3). A more detailed analysis at short reaction times reveals a bimodal pattern centered at C_4 – C_{10} and C_{25} – C_{28} (Supplementary Fig. 7). We propose two non-exclusive rationales requiring further investigations: (i) a carbenium-mediated β -scission pathway that favors near split-in-two cleavage of long chains^{24,45}; and (ii) a site-spacing effect, whereby the separation between acid sites biases scission toward these

characteristic segment lengths. Increasing N_2 pressure resulted in a slight decrease of PP conversion with negligible changes in the selectivity pattern likely due to the hindered removal of volatile products from the melt (Supplementary Table 3, Supplementary Fig. 8).

Experiments at different reaction temperatures over $W_{10}Zr_{90}$ provided new mechanistic insights. At 160 °C, near the PP_{12k} melting point (157 °C), 60% conversion and 68% selectivity to gasoline-range hydrocarbons, accompanied by relatively high fractions of heavy hydrocarbons (C_{13} – C_{30}) characterized the product distribution (Fig. 2b, Supplementary Table 3). Increasing to 180 °C drove to mildly enhanced gasoline hydrocarbons yield, consistent with selective fragmentation of long chains. At 200 °C, full conversion was achieved with a still significant C_{13} – C_{30} fraction, whereas 240 °C further enhanced gasoline selectivity to 89%, evidencing secondary cracking into shorter chains (Supplementary Fig. 9). Increasing temperature thus progressively narrowed the product distribution into the C_4 – C_{10} range.

Proton nuclear magnetic resonance, 1H NMR analysis of the products obtained at 240 °C showed that the liquids consisted predominantly of alkanes with <3% aromatics (Supplementary Fig. 10, Supplementary Table 4). The quantification of gaseous products yielded ~30% olefins and ~60% iso-alkanes (Supplementary Fig. 10, Supplementary Table 5). The abundance of saturated hydrocarbons could be explained upon elemental analysis of the solid residue, displaying a gradual decrease of its H:C molar ratio from 2.1 for virgin PP_{12k} to 1.8 after 24 h (Supplementary Fig. 11), suggesting internal hydrogen transfer within partially unreacted fragments or a dehydrogenation mechanism more prevalent on long hydrocarbon fragments requiring further investigations, supported by the presence of small amounts of H_2 in the products (0.2% yield, Supplementary Fig. 12). Evaluation at different catalyst-to-plastic ratios (1:2 and 2:1, w/w, Supplementary Fig. 13, Supplementary Table 3) confirmed C_4 – C_7 as a lower scission limit that governs selective gasoline hydrocarbons formation. Mechanistically, this can be rationalized by the structure of PP under a carbenium ion-mediated scission, since tertiary carbenium ion centers formed at methine carbons are the most stable ones²². Since the smallest fragment capable of hosting a tertiary carbenium ion is iso- C_4 , secondary depolymerization may stall around C_4 – C_5 , with C_6 – C_7 arising from neighboring scission events (and subsequent hydrogen transfer). This reasoning also supports the progressive increasing catalytic activity and narrowing of the product distribution around C_4 – C_6 products as reaction temperature increases while keeping the same reaction time (Supplementary Table 3).

A recyclability test consisting on five consecutive PP_{12k} depolymerizations over $W_{10}Zr_{90}$ at 240 °C with intermediate catalyst regeneration (calcination at 500 °C) disclosed nearly unchanged conversion and gasoline-range selectivity across the series (Fig. 2b, Supplementary Fig. 14, Supplementary Table 6). Closely aligned, STEM-HAADF images confirmed that the architecture of $W_{10}Zr_{90}$ remained seemingly intact after the recyclability test (Fig. 3). Incrementing the amount of processed plastic from 0.5 g to 3 g showed a high conversion degree (72%) while keeping the same selectivity pattern (Supplementary Table 7). $W_{10}Zr_{90}$ was further applied to diverse PP-based materials, including high-molecular-weight PP_{250k} ($M_w = 250$ kDa), centrifuge tubes, yogurt cups, and shampoo bottle caps (Fig. 2b, Supplementary Fig. 13, Supplementary Table 8). Conversion continued high (70%–93%) in all cases, with consistent gasoline selectivity (85%–88%), underscoring robustness against additives commonly presented in consumer plastics⁴⁶. The corresponding product carbon number distributions closely overlapped (Supplementary Fig. 15), suggesting a common scission mechanism at play.

Noticing that a recently reported catalyst for polyethylene depolymerization²¹ comprised of an oxidic W–Zr phase hosted by KIT6 silica and physically mixed with HZSM-5, we synthesized and evaluate its performance for PP_{12k} depolymerization under the standard

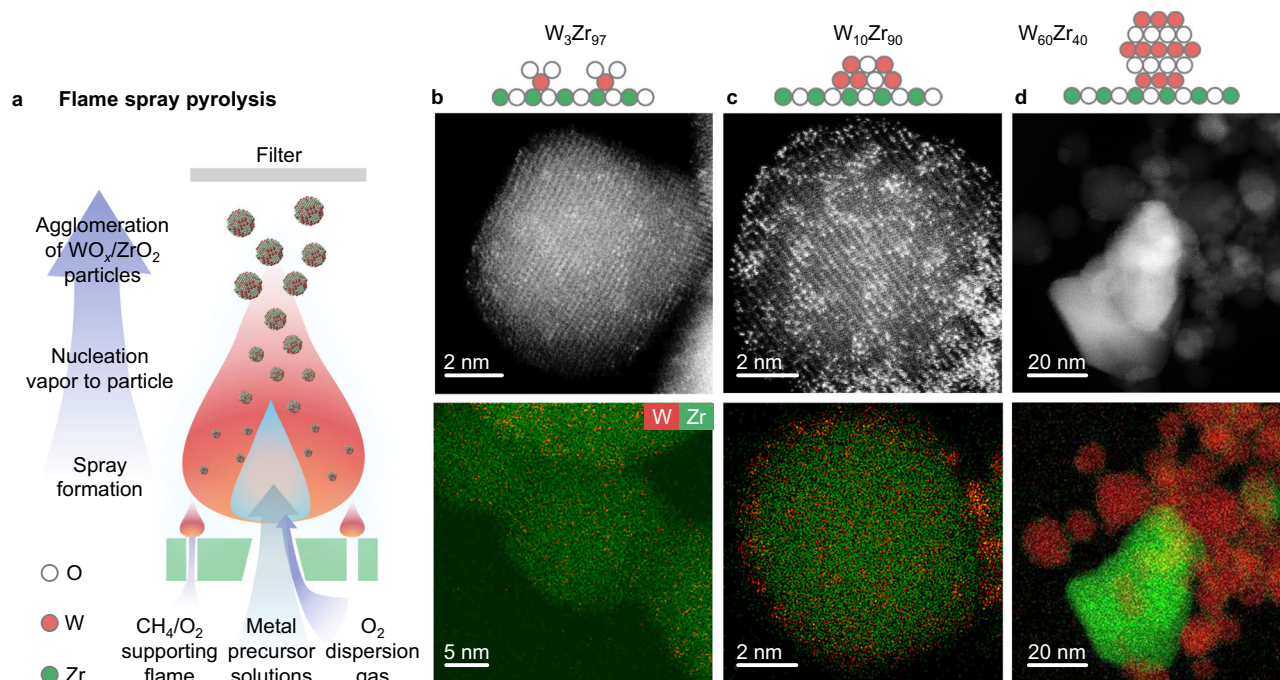


Fig. 1 | Synthesis route and nanostructure of W_aZr_b catalysts. **a** Schematic of the FSP synthesis process and representative STEM-HAADF images and elemental EDS mapping of as-prepared, **b** W_3Zr_{97} showing isolated $[WO_4]^{2-}$ species, **c** $W_{10}Zr_{90}$

showing WO_x sub-nanoclusters, and **d** $W_{60}Zr_{40}$ showing WO_3 nanoparticles on zirconia nanospheres.

conditions used in this work (Supplementary Table 9). This catalyst yielded 20% lower conversion than $W_{10}Zr_{90}$. This result highlights the high activity of W–O–Zr ensembles and suggests that efficient processing of PP may not need additional zeolite catalytic functions. Overall, these results highlighted the high performance, versatility, and stability of $W_{10}Zr_{90}$ and triggered characterization studies pursuing the nature of active sites and associated mechanisms.

Identification of W–O–Zr linkages as active sites

The distinctive architecture of the materials across the W_aZr_b family enabled clear structure–performance correlations. Starting with the analysis of electronic structure and local environments for tungsten, the W L_3 -edge X-ray absorption near-edge structures, XANES spectra identified fully oxidized W species in all cases⁴⁷, consistent with the W^{6+} valence state of the $W_{100}Zr_0$ (WO_3) standard sample (Fig. 4a). However, the sharper and more intense white lines in W_3Zr_{97} and $W_{10}Zr_{90}$ indicated distinct coordination environments from distorted octahedral in WO_3 , with enhanced W = O terminal symmetry⁴⁸. Together with STEM-HAADF (Fig. 1) and XRD (Supplementary Fig. 4) results, these data identified as predominant species isolated tungstates in W_3Zr_{97} and three-dimensional polytungstate sub-nanoclusters in $W_{10}Zr_{90}$ ^{42,49,50}. These sub-nanoclusters are likely bonded to ZrO_2 defects creating a lattice slight expansion, as suggested by the shift to lower angles of the tetragonal ZrO_2 (101) reflection as the tungsten content increased followed by reversion upon WO_3 crystallization (Supplementary Fig. 4). Fourier-transformed k^2 -weighted extended X-ray absorption fine structure, EXAFS in W L_3 -edge, together with fitting analysis (Supplementary Figs. 16–18, Supplementary Note 3), showed that the second-shell region (3.0–3.5 Å) required both W–Zr and W–W scattering contributions (Fig. 4b, Supplementary Table 10, Supplementary Fig. 17), consistent with the coexistence of interfacial W–O–Zr bridges and W–O–W environments in the catalysts. As the W content increased, the fitted W–Zr coordination number decreased, whereas the corresponding one for W–W increased, with the local W–O paths evolving from a more tetrahedral tungstate-like environment toward a more WO_3 -like coordination, consistent with XANES results (Fig. 4c

and Supplementary Note 1). Raman spectroscopy analyses of the W_aZr_b series (Fig. 4c and Supplementary Note 1) added that, for $W_{10}Zr_{90}$, ZrO_2 signals ($W_{100}Zr_0$, 150–650 cm^{-1}) nearly vanished, while the W = O modes red-shifted to 1014 cm^{-1} and 911 cm^{-1} , consistent with aggregation into sub-nanoclusters²⁹. A distinctive band at 811 cm^{-1} in $W_{10}Zr_{90}$ was attributed to W–O–Zr, consistent with the second shell W–Zr path fitted observed from W L_3 -edge EXAFS (Fig. 4b).

The abundance of W–O–Zr ensembles was evaluated via analysis of the Zr K -edge. XANES signals showed that the electronic structure and coordination environment of ZrO_2 remained essentially unchanged across W_aZr_b catalysts, providing no evidence for the formation of zirconium tungstate or solid-solution mixed oxides (Fig. 4d). While the Zr–Zr (3.6–3.7 Å) coordination numbers, CN derived from EXAFS results (Fig. 4e, Supplementary Fig. 18, Supplementary Table 11) remained consistent with a local structure resembling bulk t- ZrO_2 used as the fitting model ($CN_{Zr-Zr1} \cong 6.0$ –7.0 and $CN_{Zr-Zr2} \cong 3.5$ –4.0)^{51,52}, the fitting Zr–W (around 3.5 Å) CN showed a volcano-like dependency with the tungsten content (Supplementary Fig. 19). Accordingly, we used the ratio $CN_{Zr-W}/\Sigma CN$ as a semi-quantitative proxy for relative interface abundance, resulting in the sequence, $W_{10}Zr_{90} > W_5Zr_{95} \cong W_{30}Zr_{70} > W_3Zr_{97} > W_{40}Zr_{60} > W_{60}Zr_{40} > W_{80}Zr_{20} > W_{100}Zr_0$ (Supplementary Table 11), suggesting that $W_{10}Zr_{90}$ possessed the highest density of WO_x – ZrO_2 interfaces. $CN_{Zr-W}/\Sigma CN$ correlated linearly with PP_{12k} conversion (Fig. 4f), supporting the assignment of interfacial W–O–Zr bridges as the most relevant catalytic ensembles across the catalyst series. In this context, the W content and WO_x domain size should be regarded as governing structural parameters that modulate the abundance of such interfacial sites, as shown by the mentioned volcano-type dependency of the $CN_{Zr-W}/\Sigma CN$ ratio with tungsten content visible in the Supplementary Fig. 19.

Since W_aZr_b systems can be described as solid-acid catalysts, acidity–performance relations were explored. Temperature-programmed desorption of ammonia, NH_3 -TPD profiles and pyridine Fourier-transform infrared spectra, pyridine-FTIR analyses (Supplementary Figs. 20, 21) disclosed that, while low-W samples resemble the acidity profile of W_0Zr_{100} and high-W samples resemble that of

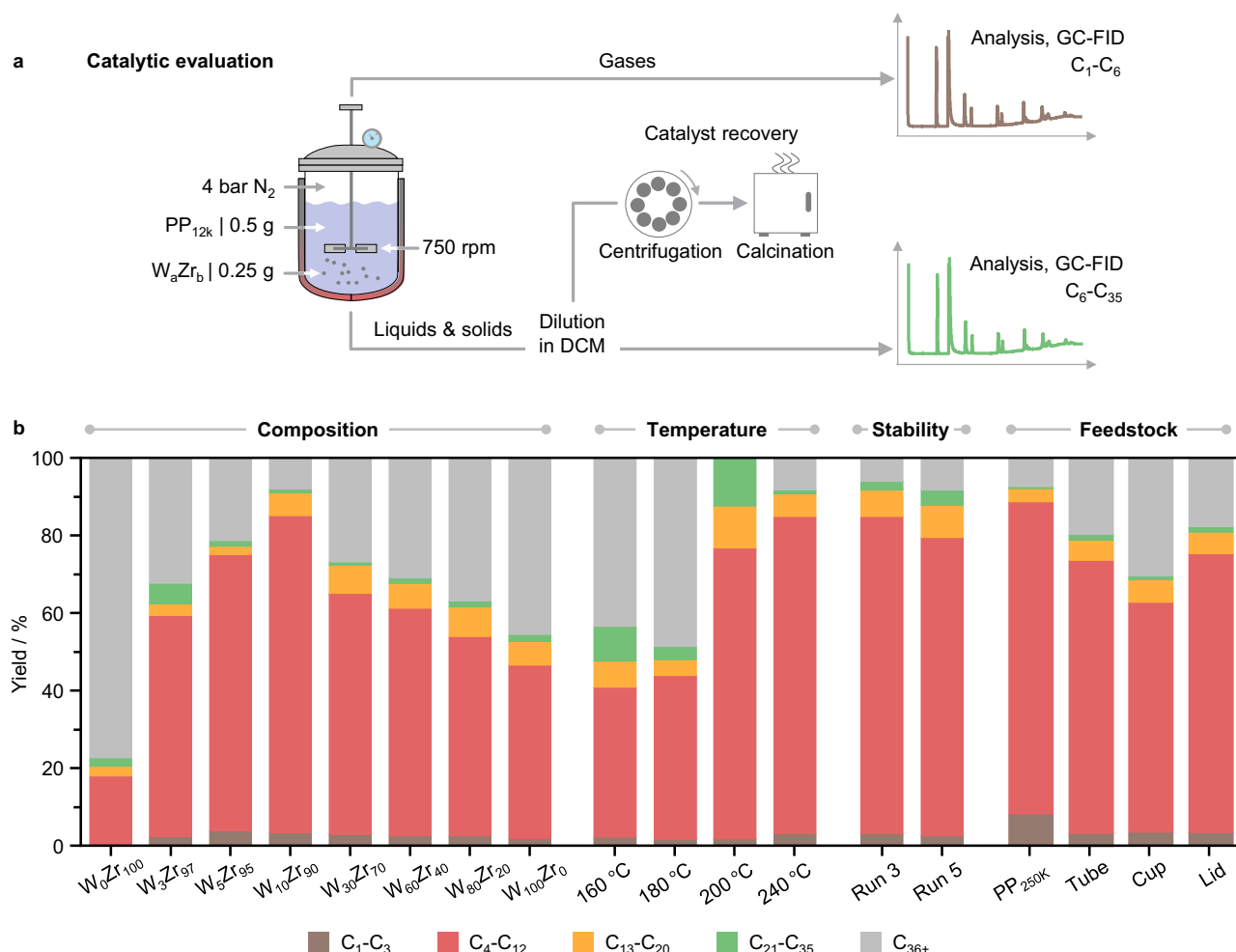


Fig. 2 | Experimental setup for polypropylene depolymerization and overview of catalytic performance over W_aZr_b catalysts. **a** Schematic of the catalyst evaluation process with indication of general conditions, product analysis, and catalyst recovery. **b** Performance for PP_{12k} depolymerization over catalysts with variable composition. $W_{10}Zr_{90}$ was selected to evaluate the influence of temperature

(160–240 °C), stability (five consecutive runs), and PP-based feedstocks (PP_{250k} , centrifuge tubes, yoghurt cups, and shampoo lids). Additional reaction conditions to those in (a): composition and stability: 240 °C, 4 h, temperature: 4 h, feedstock: 240 °C, 4–12 h. Detailed reaction conditions for each experiment are provided in Supplementary Tables 2,3,6,8. DCM stands for dichloromethane.

$W_{100}Zr_0$, respectively, $W_{10}Zr_{90}$ features fewer acid sites than W_0Zr_{100} with a higher medium-strength Brønsted/Lewis ratio (Supplementary Fig. 20). Though it is not possible to draw direct correlations between their variation and performance trends^{42,53}, first indications on the general type of acidity behind the excellent performance of $W_{10}Zr_{90}$ can be drawn. Pyridine FTIR results show that the reference $W_{10}Zr_{90}$ and W_0Zr_{100} contain an almost equal distribution of Lewis and Brønsted sites, while showing stark differences in performance (Fig. 1). The only noticeable difference is the lower magnitude of a band associated to Lewis acidity at ca. 1470 cm^{-1} for $W_{10}Zr_{90}$ (Supplementary Fig. 21). The comparison with $W_{100}Zr_0$, exhibiting a much better performance than W_0Zr_{100} , suggests that two different Lewis acid sites (1600 cm^{-1} , 1450 cm^{-1}) might be related to the catalytic cycle, but the fact that these bands are also present in W_0Zr_{100} , makes this hypothesis unlikely. Regarding NH_3 -TPD profiles, it is possible to infer that strong acid sites, only present in W_0Zr_{100} , are detrimental. The reference catalyst $W_{10}Zr_{90}$ displays a broad peak covering from weak to medium strength sites. However, W_0Zr_{100} also possesses abundant sites in this region which largely coincide with those in $W_{60}Zr_{40}$ and $W_{100}Zr_0$, showing all of them clearly different performance (Supplementary Fig. 20). The only relatively distinctive feature is thus the higher abundance of sites creating the desorption peak at ca. 283 °C in W_3Zr_{97} and $W_{10}Zr_{90}$. In summary, detailed analysis of acidity suggests

that effective depolymerization may require the absence of strong acid sites and that certain type of medium-strength sites more apparent on W_3Zr_{97} and $W_{10}Zr_{90}$ (showing the largest populations of W–O–Zr ensembles) could be related to enhanced performance. However, the ex-situ character of results provided in this section recommended to investigate the dynamics of interfacial W–O–Zr structures under operation and their mechanistic role.

Dynamics of W–O–Zr linkages under reaction conditions

Operando XANES and EXAFS built on the relevance of W–O–Zr linkages deduced from ex-situ evidence. These techniques have been rarely applied under reaction conditions in chemical plastic recycling due to their analytical challenges related to viscous, multiphase media with strong scattering and complex background signals. In a typical experiment (Fig. 5a), PP_{12k} and the catalyst were mixed (2:1, w/w) and heated in He (4 bar) at increasing temperatures while monitoring W L_3 - and Zr K -edges (Supplementary Figs. 22, 23, Supplementary Note 3). Different from W_3Zr_{97} (Supplementary Fig. 24) or $W_{60}Zr_{40}$ (Supplementary Figs. 25), the W L_3 -edge XANES spectra for $W_{10}Zr_{90}$ (Fig. 5b) showed a significant decrease in white-line intensity from room temperature to 160 °C before stabilizing, evidencing partial reduction of W species. This behavior may originate from the lower sensitivity of *operando* measurements, making that only catalysts containing

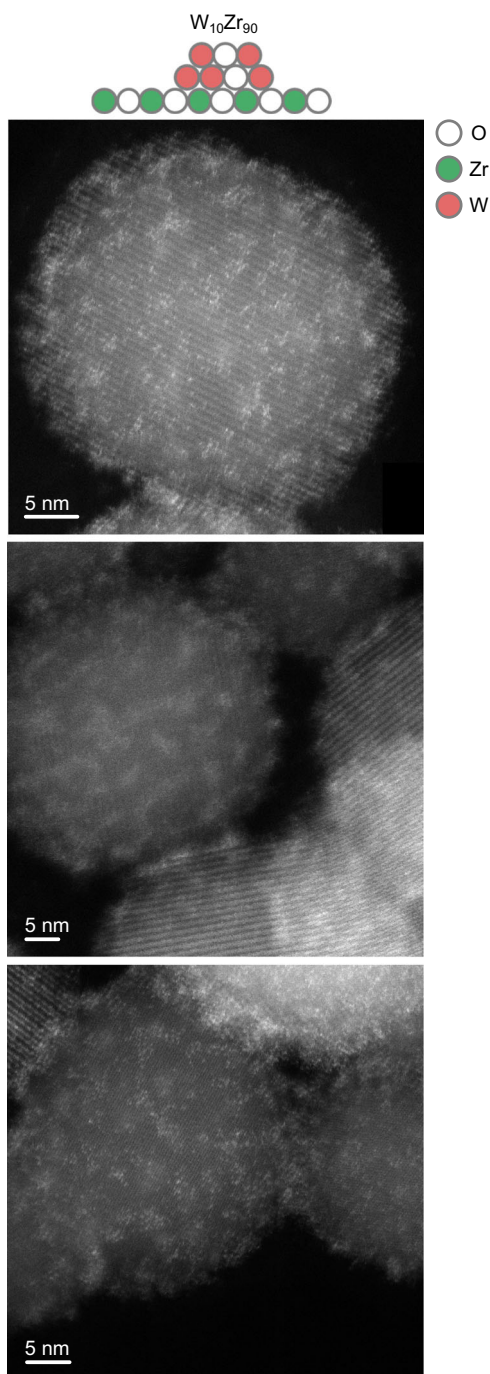


Fig. 3 | Structural stability of $W_{10}Zr_{90}$ after recyclability test. Representative STEM-HAADF images of $W_{10}Zr_{90}$ after five consecutive catalytic tests for the depolymerization of PP_{12k} . Reaction conditions: 0.25 g catalyst, 0.5 g PP_{12k} , 4 bar N_2 , 240 °C, 4 h, 750 rpm.

abundant sub-nanocluster interfaces, such as $W_{10}Zr_{90}$, display detectable reduction degrees. These results indicate the incipient activity of the catalysts even during the temperature ramping stage and that reduction may be preferentially linked to the WO_x - ZrO_2 interface predominant in sub-nanoclusters, suggesting that W-O-Zr operates as hydrogen- or electron-accepting site.

The dynamics of these linkages could be better understood from the analysis of the Zr K -edge XANES of the W_aZr_b series. In contrast to W_3Zr_{97} (Supplementary Fig. 24) and $W_{60}Zr_{40}$ (Supplementary Figs. 25), the white line intensity evolved into a distinct doublet peak (Fig. 5c) with increasing temperature, indicating substantial changes in local Zr

coordination^{54,55}. Complementary EXAFS analysis at the Zr K -edge of $W_{10}Zr_{90}$ (Figs. 5d, 5e) revealed changes in the first-shell features of both W and Zr sites. Zr K -edge EXAFS fitting showed that, although the total first-shell Zr-O coordination number remained nearly constant around eight, consistent with a tetragonal ZrO_2 local structure, the shorter Zr-O₁ subshell and the longer Zr-O₂ subshell varied from 200 °C, suggesting interfacial oxygen reorganization (Supplementary Table 12). Additionally, the Zr-W coordination number increased above 180 °C (Supplementary Fig. 26, Supplementary Table 12). Together, these changes hint to a temperature-driven restructuring at the WO_x - ZrO_2 interface, involving interfacial oxygen redistribution and the formation of additional W-O-Zr ensembles that may originate the doublet in the Zr K -edge (Fig. 5c). Of note, the emergence of new W-O-Zr linkages above 180 °C in $W_{10}Zr_{90}$ paralleled the steep increase of catalytic performance at 180 °C (Fig. 2b). *Operando* evidence thus reinstated that in-situ generated W-O-Zr structures at WO_x - ZrO_2 interfaces operate as active sites.

Main mechanistic features of polypropylene depolymerization

An analysis based on electronegativity differences between W (1.7) and Zr (1.4) already suggests that in the W-O-Zr moiety a displacement of the electronic density toward the W center may exist. The bridging oxygen would thus carry a less negative average charge, leading to milder acidity than the corresponding site in homonuclear materials. This aligns with the prevalence of acid sites with medium strength in $W_{10}Zr_{90}$ observed in Supplementary Fig. 20 and hints to distinct interactions with the PP molecule than W-O-W and Zr-O-Zr equivalents. This reasoning complements initial insights on the operation of W-O-Zr ensembles from temperature-programmed surface reaction coupled with online mass spectrometry, TPSR-MS (Fig. 6a and Supplementary Figs. 27, 28) focused on m/z 16 (CH_4 , terminal demethylation), m/z 28 (C_2H_4 , intrachain β -scission), and m/z 2 (H_2 , subsequent proton transfer). For $W_{10}Zr_{90}$, CH_4 became detectable at ca. 60 °C and reached a first maximum at around 120 °C, well below the PP_{12k} melting point. Across the W_aZr_b series, W_0Zr_{100} showed the strongest evolution of CH_4 at low temperatures with a gradual decrease with increasing W content (Supplementary Figs. 27, 28). This suggests ZrO_2 sites as the origin of low-temperature CH_4 formation. The signal for C_2H_4 initiated at ca. 120 °C over $W_{10}Zr_{90}$ and rose sharply upon polymer melting, peaking at 220 °C–240 °C. H_2 evolution started near the melting point and intensified after the C_2H_4 maximum, supporting a hydrogen-transfer mechanism, consistent with *operando* EXAFS. A two-step pathway could be thus deduced: (i) low-temperature chain-end activation by ZrO_2 sites, followed by (ii) intrachain β -scission at adjacent W-O-Zr ensembles (Supplementary Fig. 29).

Operando diffuse reflectance infrared Fourier transform spectroscopy, DRIFTS analysis at different temperatures over $W_{10}Zr_{90}$ showed comparable intensities of methyl ($-CH_3$) and methylene ($-CH_2-$) vibrations (2600 – 3200 cm^{-1}) at 60 °C, as expected from unreacted PP (Fig. 6b)^{24,56,57}. With increasing temperature, $-CH_2-$ backbone bands decreased substantially faster than $-CH_3$ side-chain/terminal bands, evidencing a predominant β -scission pathway. Close to the PP_{12k} melting point (157 °C), $-CH_2-$ bands exhibited a pronounced blue shift, while the OH stretching band of $W_{10}Zr_{90}$ simultaneously red shifted with increasing temperature (Fig. 6c). These evolutions suggest strong interactions between the polymer backbone and surface OH groups on $W_{10}Zr_{90}$ ²⁴. The dynamics and lack of shift in parallel analyses for other W_aZr_b catalysts allowed to conclude that OH groups on $W_{10}Zr_{90}$ are highly active toward β -scission (Supplementary Figs. 30, 31, Supplementary Note 2). Furthermore, the OH peak area from *operando* DRIFTS could be correlated across temperatures with the normalized C_2H_4 signal intensity from TPSR-MS (Fig. 6d). In $W_{10}Zr_{90}$, the concentration of OH species generated in-situ prior to the onset of β -scission started decreasing at ca. 120 °C, yielding a solid negative exponential correlation (Supplementary Fig. 33) and supporting the

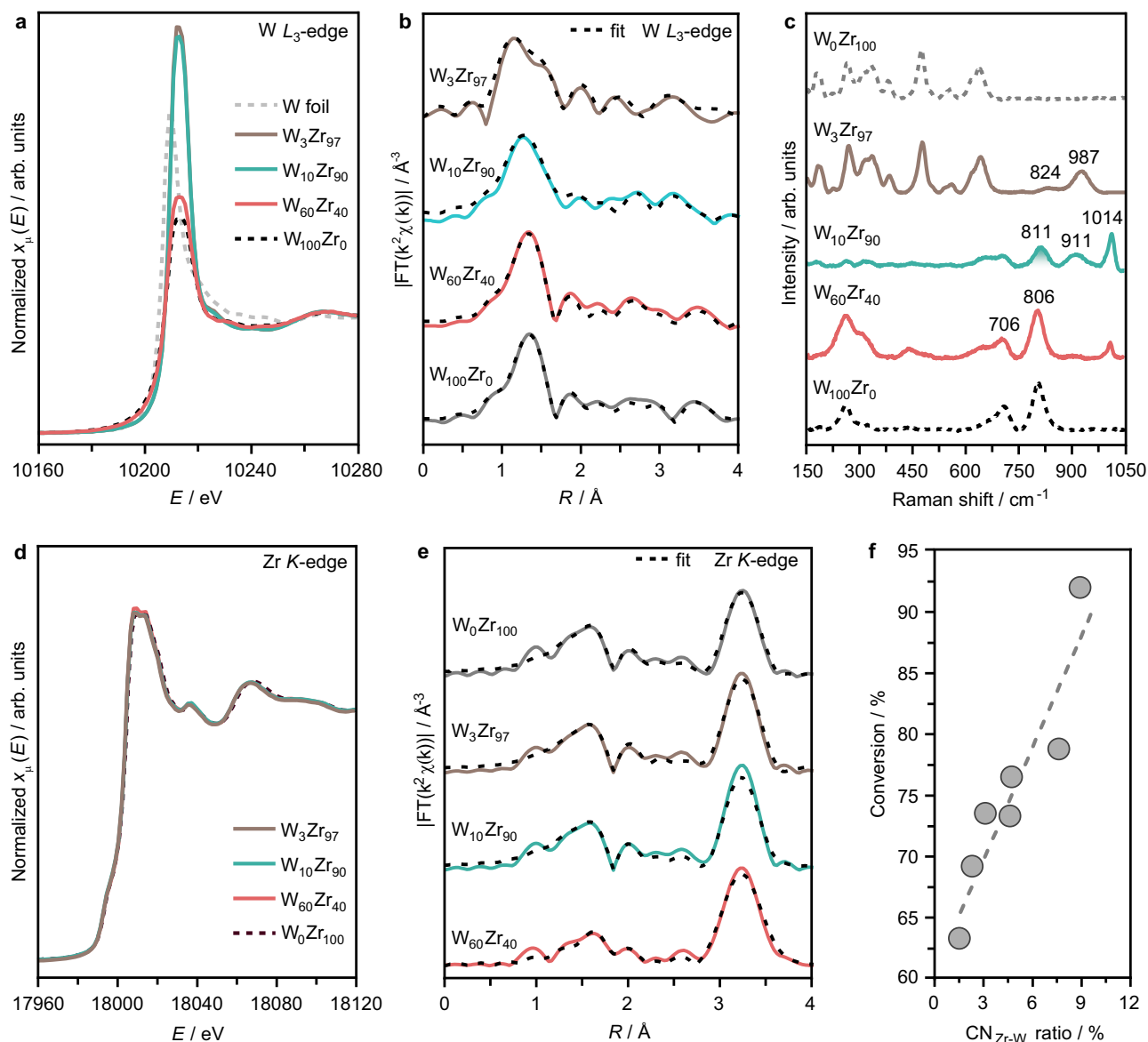


Fig. 4 | Ex-situ analysis of electronic structure and local chemical environment for representative W_xZr_{1-x} catalysts. **a** W L_{3-} edge XANES, **b** W L_{3-} edge EXAFS spectra and fittings in R space, **c** Raman spectra with peaks assignment as follows: 150–650 cm^{-1} : ZrO_2 vibrations, while the 700–1050 cm^{-1} region contained bands

from W–O–W (504–806 cm^{-1}), asymmetric (825–930 cm^{-1}) and symmetric (950–1050 cm^{-1}) W=O vibrations^{75,76}. **d** Zr K-edge XANES spectra and **e** Zr K-edge EXAFS Fourier-transform spectra and fittings in R space. **f** Correlation between PP_{12k} conversion (defined as the C₁–C₃₅ yield) and the CN_{Zr–W} ratio.

promotion of β -scission by OH groups. In contrast, the low activity of W_0Zr_{100} (Fig. 2a) with high concentration of BAS and OH groups (Supplementary Figs. 20,32) introduced the unique character of OH groups present in $W_{10}Zr_{90}$.

Furthermore, *operando* DRIFTS results are compatible with a W center first polarizing a C–H bond of the polymer chain, inducing the heterolytic cleavage^{36,58,59}, in line with the decrease in the *operando* W L_{3-} edge XANES white line (Fig. 5b). In this sequence, W may be partially reduced by accepting an electron, while the abstracted hydrogen atom is transferred to the neighboring bridging oxygen (spillover-like)^{60,61}, concomitantly generating a carbenium ion intermediate (Fig. 6e). Another supporting observation is the blue color acquired by the catalyst after reaction, consistent with the formation of tungsten bronze-type species (H_xWO_3/H_xWO_x , Supplementary Fig. 34). The decrease in OH intensity with temperature is most plausibly explained by faster hydrogen/electron reinsertion at the β -carbon, thereby completing the scission cycle and regenerating the active site (Fig. 6e). Accordingly, OH bands became barely detectable at higher

temperatures (220–240 °C) while the absence of inverted features indicated a balanced formation and consumption of OH under a rapid turnover. These W–O–Zr linkages may thus function as transient electron/hydrogen-accepting centers driving selective C–C bond scission. Of note, the same investigations applied to W_3Zr_{97} , $W_{60}Zr_{40}$, and $W_{100}Zr_0$ excluded terminal W=O and W–O–W ensembles as relevant hydrogen transfer centers (Supplementary Figs. 30–32). The returned hydrogen inserts at one end of the cleaved chain to form a new alkane, while the other end yields a new olefin, thus completing a β -scission event. Control experiments in pressurized H_2 at identical conditions showed no change in either activity or product distribution, excluding molecular hydrogen participation and further confirming the internal hydrogen transfer mechanism (Supplementary Table 13). As suggested for non-catalytic pyrolysis, the high fraction of alkanes in the product distribution could be related to the reaction of olefins with molecular H_2 ³⁴. In this case, hydrogen may originate from dehydrogenation of long fragments, as suggested by the decreasing H:C ratio in the solid residue (Supplementary Fig. 11) and the detection of

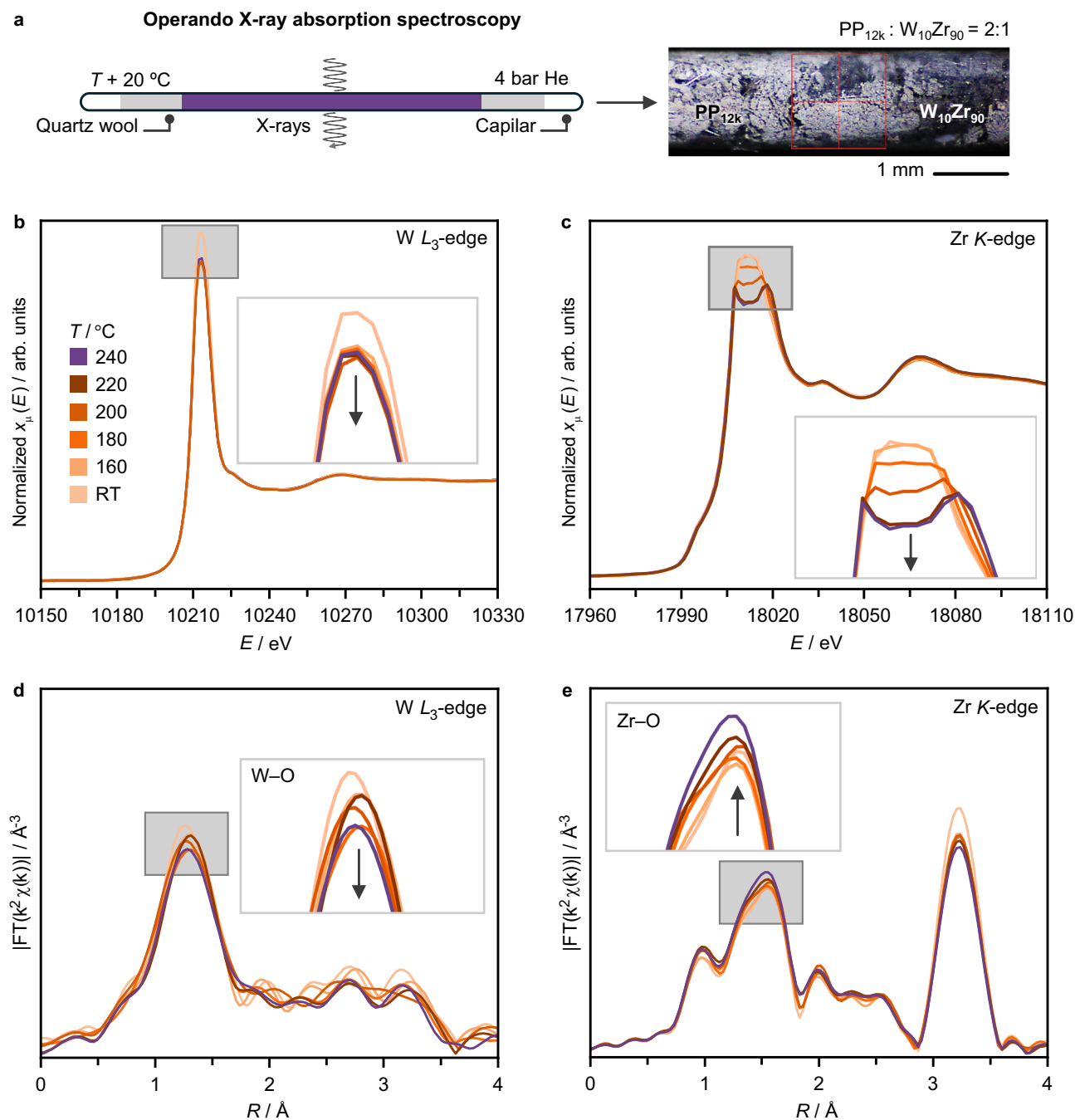


Fig. 5 | Operando analysis of electronic structure and local chemical environment for W₁₀Zr₉₀. **a** Schematic of the experimental setup for XAS analysis at different temperatures. **b** W L_3 -edge XANES, **c** Zr K-edge XANES, **d** W L_3 -edge EXAFS

(with enlarged W–O coordination region), and **e** Zr K-edge EXAFS (with enlarged Zr–O coordination region). Reaction conditions: 0.25 g catalyst, 0.5 g PP_{12k}, 4 bar He. RT stands for room temperature.

H₂ in TPSR experiments (Fig. 6a) and catalytic evaluations (Supplementary Fig. 12).

Together, these investigations converge into a coherent description of the polypropylene scission mechanism on tungstated zirconia catalysts, identifying dynamically regenerated W–O–Zr ensembles as the dominant active sites. The main features (Fig. 6e) can be described as: (i) C–H bond polarization and heterolytic cleavage: Formation of a carbenium ion intermediate by the W center. W is partially reduced by electron transfer, while the hydrogen atom generates a OH species; (ii) β -scission with hydrogen/electron reinsertion: The abstracted hydrogen atom is transferred back from the W–O–Zr linkage to the carbenium fragment, quenching it to form a new alkane and an olefinic

species. This concerted return reestablishes the W–O–Zr site and closes the catalytic cycle. Of note, additional experiments over W₁₀Zr₉₀ using HDPE as a feedstock yielded low conversion (24%, Supplementary Table 14). This result strongly suggests that the presence of tertiary and likely primary carbon atoms in PP is a relevant structural feature for the efficient operation of W–O–Zr ensembles and that catalyst design efforts may target specific polyolefins to attaining high efficiency.

Environmental and economic comparison of technologies

Life cycle and technoeconomic analyses brought positive environmental and economic prospects for the hydrogen-free PP

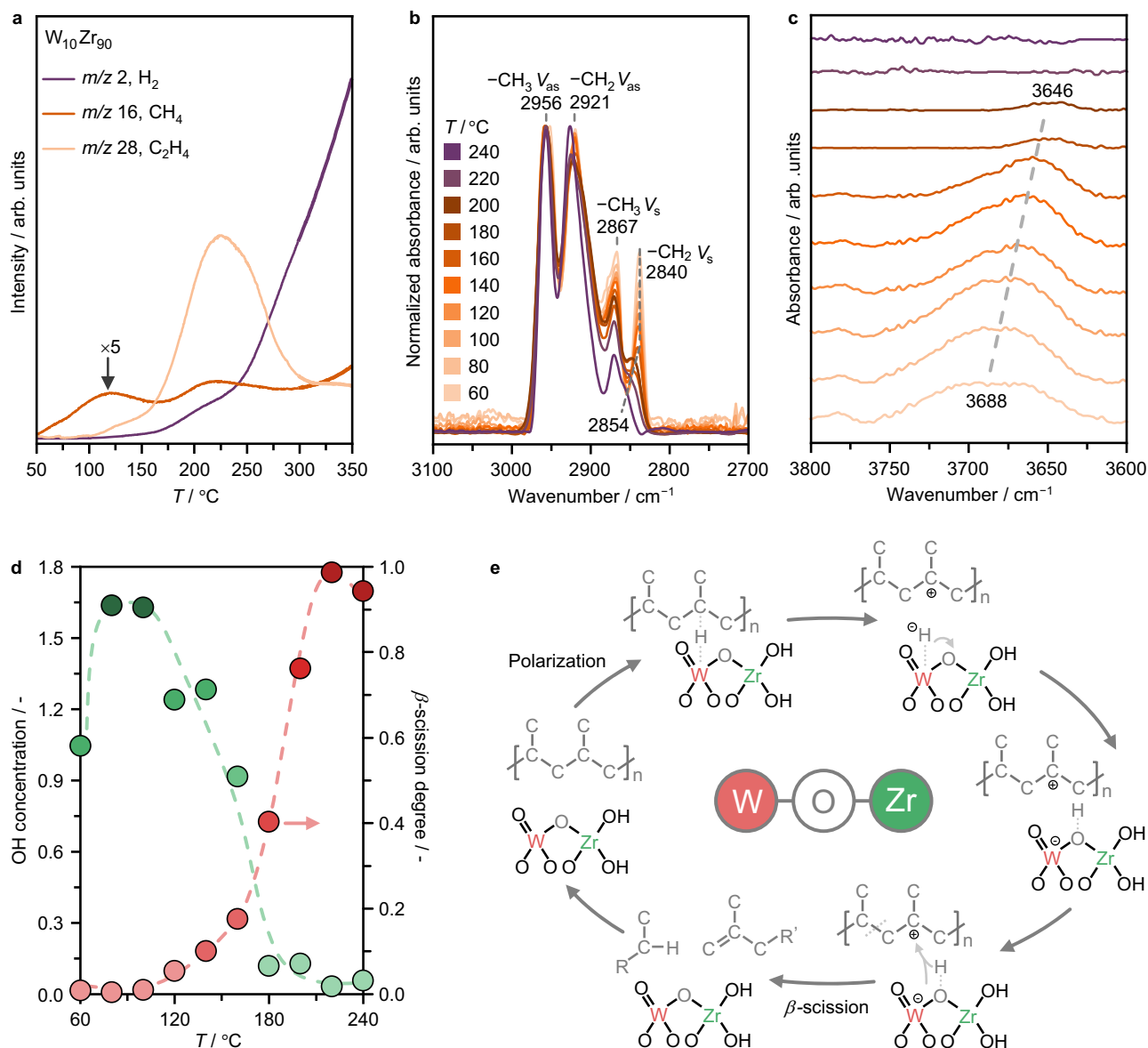


Fig. 6 | Identification of key mechanistic features for the depolymerization of PP over $W_{10}Zr_{90}$. **a** TPSR-MS profiles, *operando* DRIFTS evolution of **b** PP C–H stretching, and **c** OH stretching. **d** Correlation between OH concentration (from

integrated DRIFTS band at $3600\text{--}3770\text{ cm}^{-1}$) and normalized C_2H_4 MS intensity (m/z 28) for the depolymerization of PP_{12k} over $W_{10}Zr_{90}$. **e** Proposed main reaction pathway.

chemical recycling strategy over $W_{10}Zr_{90}$. We simulated an industrial plant processing 20 t h^{-1} of consumer goods (centrifuge tubes, yogurt cups, shampoo lids, catalytic performance in Fig. 2b) integrating the reaction stage with additional units for stream conditioning, catalyst regeneration, and product separation (Fig. 7a, detailed information of modeling in Supplementary Fig. 35 and Methods section). The environmental impact was assessed using the global warming potential, GWP indicator over the full life cycle.

The business-as-usual, BAU scenario reflects the current situation, where PP-based consumer goods are produced from fossil feedstocks and, at the end of life, managed through conventional processes (34% landfilling and 66% incineration), while fuels are also obtained from fossil resources in a refinery (Fig. 7b). The hydrogen-free recycling route leads to an average 35% reduction in GWP compared to the BAU scenario (Fig. 7c and Supplementary Figs. 36–38). Depolymerization of all consumer goods show similar

performance, with combined propylene production, collection, sorting, and transport being the dominant GWP contributor (ca. 80%), in which the production stage via fossil-based energy-intensive processes, most notably the steam cracking of naphtha, contributing with the largest share (Supplementary Fig. 38). The economic analysis indicates a favorable outlook (Fig. 7d), with profit margins of $0.5\text{--}0.7\text{ \$ per kg}$ of product when comparing production costs and expected product revenues at market prices of the product portfolio. This result is driven by high yields toward liquid hydrocarbons, mild reaction conditions, and the use of an inexpensive catalyst. The largest cost contribution stems from the acquisition of waste PP-based items (70%), which is traded as a commodity requiring collection and processing prior to sale. A sensitivity assessment predicted that, even under unfavorable raw materials and energy prices, market fluctuations, and considering life cycle inventory data uncertainty, the process exhibits consistent profits and CO_2 savings across (Supplementary Fig. 38).

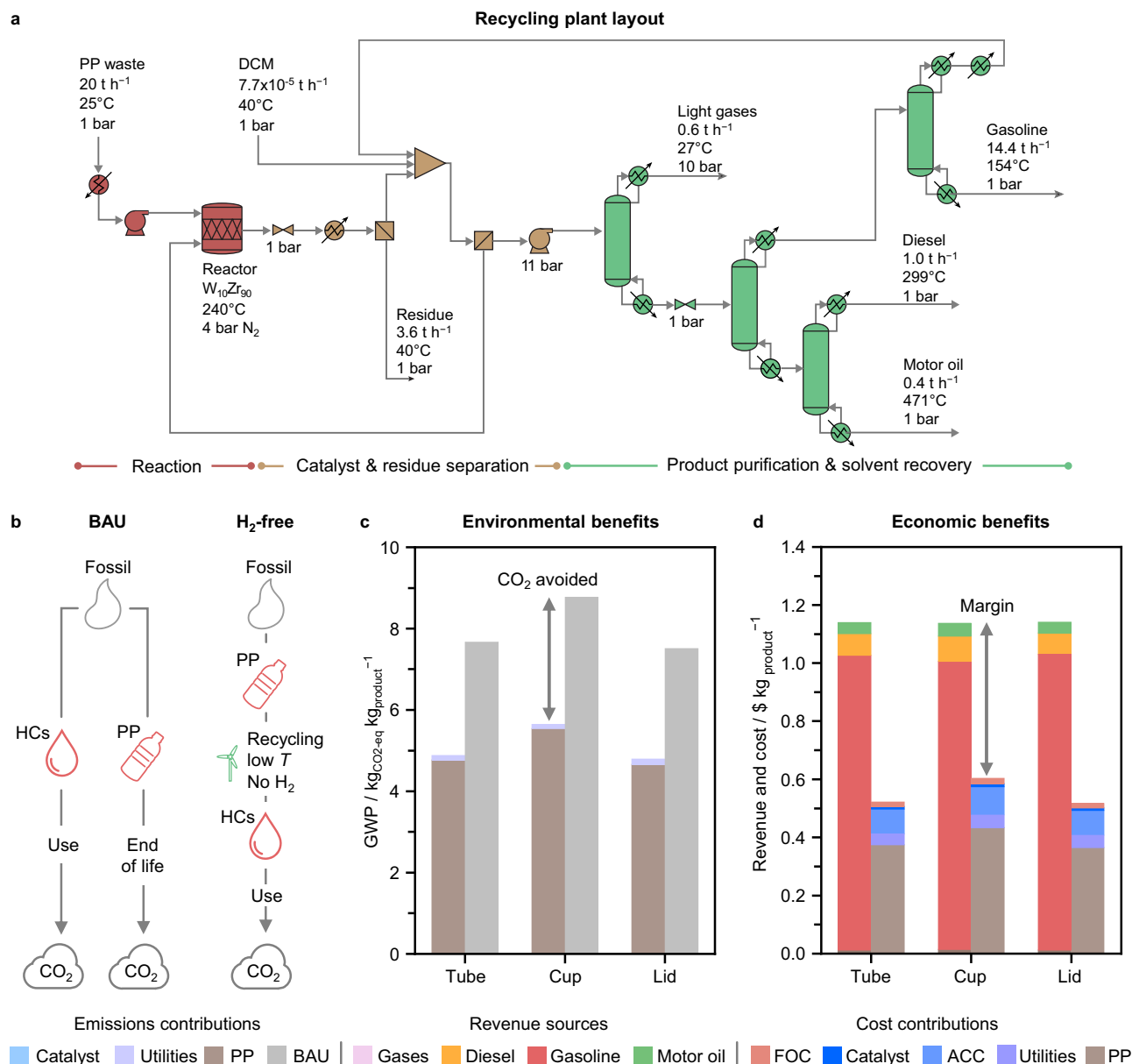


Fig. 7 | Analysis of environmental and economic performance for the depolymerization of PP-based consumer goods over $W_{10}Zr_{90}$. **a** Simplified process flowsheet for the chemical recycling process considering catalytic performance provided in Supplementary Table 8. The values in the figure correspond to the shampoo lids scenario. **b** General schemes of life cycles for the hydrogen-free depolymerization of polyolefins and for the business-as-usual, BAU production of polyolefins and liquid hydrocarbons, HC's. **c** Global warming potential versus

equivalent business-as-usual fossil production and **d** production cost and revenues from product selling at market prices, with margin indication of their respective product portfolio. Detailed breakdowns and system boundaries are available in Supplementary Fig. 38 and Supplementary Fig. 39, respectively. DCM stands for dichloromethane, FOC stands for fixed operating costs and ACC for annualized capital cost.

This framework was extended to other polyolefin chemical recycling technologies requiring similar temperatures, inspired by recent works generalizing sustainability analysis of hydrotreatments⁶². The analysis included representative hydrotreatment reports including hydrogenolysis and hydrocracking (Supplementary Fig. 39) addressing consumer goods with detailed product chain length ranges (plant layout in Supplementary Fig. 40). This strategy enabled a harmonized comparison including uncertainty analysis depicted in Fig. 8 as a technology sustainability map. Overall, chemical recycling of polyolefins is found to be sustainable in both the environmental and economic dimensions with nuances even under less favorable scenarios. Hydrogen-free processing of PP-based consumer goods over $W_{10}Zr_{90}$ sits at the forefront zone. As for hydrotreatments, reports form a

distinct and broader performance region due to their dependence on green hydrogen (sourced from wind and solar water splitting in our analysis), frequent use of noble-metal catalysts in hydrogenolysis (e.g., Ru, Pt), and still mild yields of highly valuable lubricant hydrocarbons. Uncertainty bars account for raw materials and energy prices, market fluctuations, and life cycle inventory data ranges, making apparent the large variability that sustainability performance is subject to and highlighting the relative comparison across technologies as the main result of this analysis. All in all, chemical recycling routes of polyolefin waste offer a marked reduction in greenhouse gas emissions compared to conventional end-of-life options, while generating product portfolios that, depending on the specific technology and its performance, may offer favorable economic prospects.

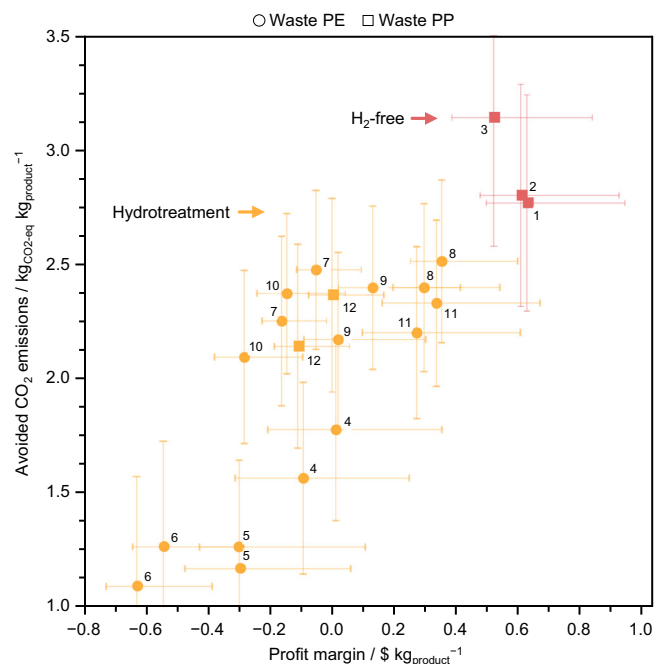


Fig. 8 | Environmental and economic performance map for polyolefin chemical recycling technologies at mild conditions. Hydrotreatment and hydrogen-free routes are compared in terms of profit margin ($\$/\text{kg}_{\text{product}}^{-1}$) and net GWP reduction over the BAU system ($\text{kg}_{\text{CO}_2\text{-eq}}/\text{kg}_{\text{product}}^{-1}$). Number labels, performance, and catalytic materials for each report are provided in Supplementary Tables 22, 23. The uncertainty zone for each point (vertical and horizontal lines) is computed considering uncertainty in fuel and material prices as described in Supplementary Table 18 and a $\pm 25\%$ GWP impact change in the chemical recycling routes. The hydrotreatment routes are evaluated using green hydrogen produced with electricity from either solar PV or offshore wind.

Discussion

This work establishes a hydrogen- and solvent-free catalytic route for polypropylene depolymerization at mild temperatures. Flame-synthesized tungstated zirconia catalysts unify high activity and selectivity toward gasoline hydrocarbons (>80% yield) with well-defined structures, enabling identification of W–O–Zr ensembles as active sites. *Operando* spectroscopies reveal the formation of these interfacial linkages under reaction and their operation as transient electron–hydrogen accepting sites that activate C–H bonds and guide selective C–C scission through internal hydrogen transfer. Process modeling, life cycle assessment, and a harmonized comparison across chemical recycling technologies including hydrotreatments show that this pathway already sits at the forefront of environmental and economic performance. By linking molecular-level catalyst design to validated system-scale benefits, these findings suggest interfacial M–O–M' ensembles (M=metal) as a design platform for advancing catalytic polyolefin waste management.

Methods

Catalyst synthesis

Tungstated zirconia catalysts with variable composition (denoted as W_aZr_b , where a and b represent the mol% of tungsten and zirconium, respectively, listed in Supplementary Table 1) for the recycling of polypropylene was synthesized via flame spray pyrolysis (FSP) using 0.10 mol kg^{-1} stock solutions of tungsten hexacarbonyl ($\text{W}(\text{CO})_6$, Acros; 99%) and zirconium(IV) 2-ethylhexanoate ($\text{Zr}(\text{2-ethylhexanoate})_4$, abcr; 97%) in tetrahydrofuran (THF, Sigma-Aldrich, $\geq 99.9\%$) as precursors, by mixing the stock solutions in the desired ratios and sonicating for 30 min. The FSP setup has been described elsewhere in detail⁶³. In brief, the liquid metal precursor was fed

through a spray nozzle (0.4 mm) by a micro-annular gear pump (HNPM; mzt-2900) at a rate of $10.0\text{ cm}^3\text{ min}^{-1}$. The spray was dispersed by oxygen ($5.0\text{ dm}^3\text{ min}^{-1}$; 1.5 bar overpressure) and ignited by a premixed pilot flame of CH_4 ($1.2\text{ dm}^3\text{ min}^{-1}$) and O_2 ($2.4\text{ dm}^3\text{ min}^{-1}$) surrounding the nozzle reaching $2000\text{ }^\circ\text{C}$ – $3000\text{ }^\circ\text{C}$ based on adiabatic calculations⁶³. The gas flow through the reactor was maintained at $10\text{--}20\text{ m}^3\text{ h}^{-1}$ by a vacuum pump (Mink MV 0040 D; Busch) using ambient air passing through a HEPA filter. The obtained catalyst particles were separated from the gas stream using glass fiber filters (Hahnemühle, GF6, diameter 257 mm).

The WZr-KIT6-HZSM-5 catalyst was synthesized following a previously reported method²⁴. Briefly, 3.7 g of Pluronic P123 (Sigma Aldrich; M_n - 5800 Da), 6.8 g of hydrochloric acid solution (Sigma Aldrich; 37 wt%), and 3.7 g of *n*-butanol (VWR international AG; 99.5%) were dissolved in 130 g of deionized water at $35\text{ }^\circ\text{C}$. The mixture was stirred for 1 h to ensure complete dissolution. Subsequently, 8.0 g of tetraethyl orthosilicate (TEOS; Sigma Aldrich; 98%), 1.24 g of zirconyl chloride octahydrate (Sigma Aldrich; 98%), and 0.95 g of ammonium metatungstate (ABCR; 99.9%) were added. The resulting mixture was stirred at $35\text{ }^\circ\text{C}$ for at least 24 h, after which it was transferred into four 50 mL stainless steel autoclaves and subjected to hydrothermal treatment at $100\text{ }^\circ\text{C}$ for 24 h. The solid product was then recovered by filtration, dried at $100\text{ }^\circ\text{C}$, and finally calcined in air at $550\text{ }^\circ\text{C}$ for 5 h. HZSM-5 was prepared from calcination of ammonium ZSM-5 (Zeolyst International) in air at $550\text{ }^\circ\text{C}$ for 10 h and then physically mixed with WZr-KIT6 in a four to one mass ratio, respectively.

Ex-situ catalyst characterization

X-ray fluorescence, XRF spectroscopy was performed using a Rigaku ZSX Primus IV spectrometer equipped with a 4 kW Rh source and LiF (200), Ge, PET, and RX26 analyzing crystal detectors.

Powder X-ray diffraction, XRD measurements were conducted in a Rigaku SmartLab diffractometer, using $\text{Cu K}\alpha$ radiation ($\lambda = 1.541\text{ \AA}$). The data was recorded in the 2θ range of 10° – 90° with an angular step size of 0.025° and a counting time of 1.5 s per step.

Temperature-programmed surface reaction, TPSR experiments were performed in a Micromeritics Autochem II 2920 unit equipped with a thermal conductivity detector, TCD and a Pfeiffer Vacuum Omni Star mass spectrometer, MS. The catalyst (90 mg) was thoroughly mixed with polypropylene with weight average molecular weight, M_w - 12 kDa, (10 mg, Sigma-Aldrich), denoted as PP_{12k}, and loaded into a U-shaped quartz micro-reactor followed by drying in He ($40\text{ }^\circ\text{C}$, 1 h). The sample was subsequently heated in flowing He ($20\text{ cm}^3\text{ min}^{-1}$) to $600\text{ }^\circ\text{C}$ at a rate of $5\text{ }^\circ\text{C min}^{-1}$.

Temperature-programmed desorption of ammonia, NH_3 -TPD was performed in a Micromeritics Autochem II 2920 unit equipped with a thermal conductivity detector and online MS. Prior to NH_3 -TPD measurements, samples (100 mg) were loaded into a quartz tube, dried under flowing He at $150\text{ }^\circ\text{C}$ for 1 h ($20\text{ cm}^3\text{ min}^{-1}$, heating rate $10\text{ }^\circ\text{C min}^{-1}$), and then cooled down to $40\text{ }^\circ\text{C}$ ($20\text{ }^\circ\text{C min}^{-1}$). After 10 min, the sample was exposed to flowing NH_3 in He (10 vol%, $20\text{ cm}^3\text{ min}^{-1}$) for 1 h. The initial desorption of weakly bound NH_3 was conducted at $60\text{ }^\circ\text{C}$ under flowing He ($20\text{ cm}^3\text{ min}^{-1}$) for 20 min. Temperature-programmed desorption was then performed under flowing He ($20\text{ cm}^3\text{ min}^{-1}$) in the $40\text{ }^\circ\text{C}$ – $800\text{ }^\circ\text{C}$ range ($10\text{ }^\circ\text{C min}^{-1}$), with NH_3 evolution monitored by a TCD and MS.

Raman spectroscopy was performed in a Horiba LabRAM HR Evolution UV–vis–NIR confocal Raman system using a Cobolt Samba Nd/YAG laser with a wavelength of 532 nm, a power of 5.7 mW and a $50\times$ Olympus LMPlanFLN objective. Spectra were collected with an acquisition time of 10 s and an accumulation number of 5.

Pyridine Fourier-transform infrared spectra were recorded using a Bruker Optics Vertex 70 spectrometer. For each measurement, the catalyst (200 mg) was pressed into a thin wafer and placed in a custom-made quartz reaction cell with KBr windows, connected to a molecular

pump and capable of in situ heating. The sample was evacuated at 300 °C under vacuum for 120 min, cooled to 50 °C, and exposed to pyridine for 10 min. After evacuation for 30 min to remove physisorbed species, IR spectra were collected at 50 °C intervals in the 150 °C–350 °C range.

High-angle annular dark-field scanning transmission electron microscopy, HAADF-STEM and energy-dispersive X-ray spectroscopy, EDX data were acquired using an aberration-corrected JEM-ARM300F microscope (GrandARM, JEOL) operated at 300 kV or a Talos F200X (Thermo Fisher Scientific).

Ex-situ X-ray absorption spectroscopy, XAS was conducted at the Swiss Norwegian beamline, SNBL, BM31 of the European Synchrotron Radiation Facility (ESRF)⁶⁴. The X-ray beam was collimated using a double-crystal liquid N₂ cooled Si (111) monochromator. For the absolute energy calibration for XAS measurements, a W foil and a Zr foil were measured using N₂-filled ionization chambers for optimal absorption levels. Continuous scanning was performed for both W-*L*₃ and Zr-*K* edges at a step size of 0.5 eV. The beam size was set to 3 × 0.2 mm (horizontally × vertically). The resulting spectra were energy-calibrated, background-corrected and normalized using the Athena program from the Demeter software suite. *k*²-weighted extended X-ray absorption fine structure, EXAFS spectra were fitted in the optimal *k*- and *R*-windows using the Artemis program.

Operando catalyst characterization

Operando diffuse reflectance infrared Fourier transform spectroscopy, DRIFTS was applied using a Bruker Invenio S spectrometer following this procedure: (i) the catalyst was mixed with PP_{12k} in a 1:2 mass ratio and loaded in the cell, (ii) the sample was flushed with He and pressurized to 4 bar, (iii) background data was collected once the pressure and signal had stabilized, and measurements were performed starting at 60 °C, increasing by 20 °C every 0.5 h until reaching 240 °C, which was then maintained for 2 h; (iv) the system was naturally cooled down to room temperature. Spectra were recorded at 0.5 min intervals with a resolution of 4 cm⁻¹.

Operando XAS experiments were conducted at the same beamline of BM31, SNBL, ESRF. The X-ray beam was collimated using a double-crystal liquid N₂ cooled Si (111) monochromator⁶⁴. For the absolute energy calibration for XAS measurements, a W foil and a Zr foil were measured, using N₂-filled ionization chambers for optimal absorption levels. Continuous scanning was performed for both W *L*₃- and Zr *K*-edges at a step size of 0.5 eV. The beam size was set to 3 × 0.2 mm (horizontally × vertically). The catalyst and the plastic were placed between two plugs of quartz wool in a quartz capillary reactor cell (1.5 mm outer diameter, 0.01 mm wall thickness). The experiment procedure was as follows: (i) the catalyst was mixed with PP_{12k} in a 1:2 mass ratio and loaded in the cell; (ii) the sample was flushed with He and pressurized to 4 bar; (iii) data collection was initiated at room temperature, where consecutive spectra were acquired five times at both the W *L*₃-edge and Zr *K*-edge; (iv) the temperature was then increased to 160 °C, followed by stepwise heating in 20 °C increments to 240 °C. At each target temperature, the system was stabilized for 30 min, and consecutive spectra were again collected five times at both the W *L*₃-edge and Zr *K*-edge. The high-purity gases were introduced by a set of Bronkhorst digital mass flow controllers, and the outcome was monitored on-line via a Pfeiffer Vacuum Omni Star GSD 3200 MS with QUADERA software. The resulting spectrum was energy-calibrated, background-corrected and normalized using the Athena program from the Demeter software suite. *k*²-weighted EXAFS spectra were fitted in the optimal *k*- and *R*-windows using the Artemis program.

Catalyst evaluation

Catalytic depolymerization of polypropylene under hydrogen-free conditions was performed in a parallel pressurized batch reactor setup (BuchiGlasUster, Switzerland) consisting of three 50 cm³ (110 cm³ for

gas volume) stainless steel reactors. Each reactor was equipped with an electrical heating jacket, active cooling unit (chilled water system), mechanical stirring, temperature/pressure control systems and gas sampling lines. Typically, plastics (0.5 g) including virgin polypropylene PP_{12k} (also used for mechanistic studies due to its lower molten viscosity amenable to FTIR and TPSR operation), PP_{250k} (*M*_w = 250 kDa, Sigma Aldrich), HDPE_{100k} (*M*_w = 100 kDa, Sigma Aldrich) and consumer goods based on PP (centrifuge tube, yogurt bottle, and shampoo lid; washed and cut into pieces before using with catalysts (0.25 g) were added to a glass inset and the combined weight was recorded, *m*_{before}. The inset was then placed inside the reactor. Prior to starting the experiment, the reactor was flushed with N₂ before being pressurized to the desired pressure. The reaction parameters and individual steps were programmed into the Systag Flexsys software before starting the experiment. Temperature, pressure and stirring torque were recorded during the reaction. The reaction mixture was heated and mechanically stirred for a set reaction time after having reached the set temperature (4 bar, 240 °C, 750 rpm and 4 h, respectively, unless otherwise specified). PID control parameters were chosen to avoid temperature overshooting. This implied slightly different temperature ramping processes across different conditions, averaging 3 °C min⁻¹. After reaction, the reactor was cooled to ambient temperature with circulating chilled water.

Product analysis

Gaseous products were collected from the headspace of the reactor by connecting a sampling cylinder and analyzed using a gas chromatographer Agilent 8860 GC System equipped with an Agilent J&W Poraplot Q column (25 m × 0.53 mm × 20 μm) and a flame ionizing detector, FID. H₂ was used as the carrier gas and a heating ramp (35 °C–300 °C, 5 °C min⁻¹) was applied while the inlet and FID were held at fixed temperatures of 300 °C and 200 °C, respectively. GC FID calibration for C₁ to C₆ products was performed using a standard refinery mixture (Agilent P/N: 5080-8755, Agilent Technologies Inc.). The glass inset was removed from the reactor and weighed (*m*_{after}) to determine the total amount of gas (*m*_{gas}) formed with Eq. 1.

$$m_{\text{gas}} = m_{\text{before}} - m_{\text{after}} \quad (1)$$

The liquid and solid products remaining inside the inset were dissolved in dichloromethane (10 cm³), sonicated (40 °C, 30 min) and filtered using a syringe. The weight of filter and syringe were measured before and after their use in completely dry conditions, with the difference in mass corresponding to the amount of solid residue, *m*_{residue}. The amount of liquid products, *m*_{liquid} was calculated with Eq. 2 by subtracting the gas product and solid residue from the amount of substrate, *m*_{feedstock} used. This procedure operates on the assumption of no loss of material during operation and thus on a closed carbon balance.

$$m_{\text{liquid}} = m_{\text{feedstock}} - m_{\text{gas}} - m_{\text{residue}} \quad (2)$$

The filtered liquid samples were analyzed in an HP Agilent 6890 GC (Agilent Technologies, Inc., USA) equipped with an HP DB-5 HT column (15 m × 0.25 mm × 0.10 μm) to determine the distribution of products. A heating ramp was applied (40 °C–375 °C, 4 °C min⁻¹), while the FID was held at a fixed temperature of 340 °C. Initial and final hold times were set at 2 and 10 min, respectively. GC FID calibration was performed using a mixture of self-prepared alkanes solution with carbon numbers of 6, 7, 8, 10, 12, 16, 20.

The yields expressed as percentage of gas phase, Y_{gas} and liquid, Y_{liquid} products were calculated with Eqs. 3 and 4:

$$Y_{\text{gas}} = \frac{m_{\text{gas}}}{m_{\text{feedstock}}} \cdot 100 \quad (3)$$

$$Y_{\text{liquid}} = \frac{m_{\text{liquid}}}{m_{\text{feedstock}}} \cdot 100 \quad (4)$$

The overall conversion, X was calculated according to Eq. 5:

$$X = Y_{\text{gas}} + Y_{\text{liquid}} \quad (5)$$

The yield expressed in percentage for each carbon range, Y_c was calculated with Eq. 6:

$$Y_c = \frac{\sum_{i=1}^j [C_i] \cdot Y}{\sum_{i=1}^{36} [C_i]} \cdot 100 \quad (6)$$

where Y_c is the yield of hydrocarbons with a carbon-number range from i to j ; $[C_i]$ is the concentration of the i^{th} hydrocarbon; Y is the overall yield towards gas or liquid products, respectively.

The gasoline range selectivity expressed in percentage was calculated with Eq. 7:

$$S_{\text{gasoline}} = \frac{\sum_{i=4}^{12} [Y_c]}{X} \cdot 100 \quad (7)$$

where S_{gasoline} is the selectivity of gasoline with the carbon-number range from 4 to 12.

For proton nuclear magnetic resonance, ^1H NMR spectroscopy analysis of the liquid products, the sample was dried at room temperature using a rotary evaporator to remove the dichloromethane solvent and then dissolved in dichloromethane- d_2 to a concentration of 100 mg mL $^{-1}$. The ^1H NMR spectra were recorded on a 500 MHz Bruker Ultrashield spectrometer.

The H and C contents of the solid residues were determined using a LECO TrueSpec Micro elemental analyzer. Samples were dried under vacuum at 80 °C overnight before measurement. Typically, the sample (~1 mg) was combusted in pure O_2 at ~950 °C, and the evolved gases were quantified by infrared spectrometry (C and H) detection. The instrument was calibrated with acetanilide, and each value represents the average of at least three replicates.

Process modeling of polypropylene chemical recycling

Polypropylene chemical recycling was assessed based on experimental conversion and selectivity obtained using the $\text{W}_{10}\text{Zr}_{90}$ catalyst. A process model was developed in Aspen Plus v15 using the Peng-Robinson property method, suitable for non- or mildly polar mixtures containing hydrocarbons and light gases. Waste PP was modeled as a polyolefin with a molecular weight of 250 kDa and an average input flow of 20 t h $^{-1}$, following assumptions from similar works^{62,65}.

The process flowsheet, shown in Supplementary Fig. 35, considers the reactor stage as well as detailed downstream separation blocks. PP (25 °C, 1.013 bar) is fed to the process, heated (240 °C) in H1, and pumped (4 bar) in P1 to reaction conditions. The reactor (R1) was modeled as a RYield reactor block operating at 240 °C and 4 bar, with the tungstated zirconia catalyst enabling the conversion and selectivity described in Supplementary Table 8 for the same conditions. The outlet stream is first cooled to 40 °C (C1) and depressurized to 1.013 bar (V1) before passing through a decanter centrifuge that removes the solid residue. The liquid is then mixed with dichloromethane (DCM) and filtered, thereby regenerating the catalyst, which is recycled to reactor R1. It is assumed that total catalyst replacement occurs after 1000 h of operation. The resulting stream, containing the

product mixture and solvent, is sent to the downstream separation section. The four main products (i.e., light gases, gasoline, diesel, and motor oil) are purified to high molar purity in four distillation columns (D1–D4) modeled with the RADFRAC rigorous model. The dichloromethane, DCM solvent is recovered in the condenser of column D3, recycled to mixer M1, and then returned to the catalyst regeneration unit. To compensate for process losses, particularly during product recovery, a DCM make-up stream is required. Finally, heat integration was performed in Aspen Energy Analyzer v15 to design a heat exchanger network that optimally integrates the hot and cold process streams to reduce utilities consumption.

Technoeconomic and life cycle assessment

Mass and energy balance data were obtained from the corresponding simulation to calculate costs and build life cycle inventories, LCI⁶⁶. The technoeconomic assessment, TEA determines the total annualized cost, TAC of producing the end-product portfolio and was calculated using the cost factors and correlations from Towler and Sinnott⁶⁷. The TAC is calculated as a function of the variable operating cost, VOC, the fixed operating cost, FOC, and the annualized capital cost, ACC.

The ACC and FOC were also derived using cost correlations from Towler and Sinnott⁶⁷. The general assumptions of the plant are described in Supplementary Table 15. The values used to estimate capital costs are provided in Supplementary Table 16 for each equipment type. The inside battery limits cost, ISBL is calculated based on the purchasing cost, C and typical installation factors for ‘fluids-solids’ processes⁶⁷. The ISBL is scaled considering additional costs (offsite, engineering, contingencies) to obtain the total fixed capital cost, FCC. This term is then annualized to provide the annual capital charge ratio, accounting for accumulated interest over the plant’s lifetime.

The FOC is determined from the FCC and the cost of labor, CL. It comprises expenses linked to operating labor, supervision, salary overheads, and maintenance, among others⁶⁷. The CL corresponds to five operators (three shift positions plus two operators for the catalyst regeneration and residue section) with an average yearly salary of USD 60'000 per person. All TEA values are expressed in USD 2024. Cost data were adjusted to the selected year using the Chemical Engineering Plant Cost Index, CEPCI when not directly available. The cost parameters used in the VOC are shown in Supplementary Table 17, while the market prices considered for the revenues are presented in Supplementary Table 18.

The LCA followed the methodology described in the ISO 14040/14044^{68,69} standards with the Ecoinvent 3.10 database⁷⁰ as background system and the outputs of the process simulations as foreground data. The global warming potential, GWP of the different scenarios was calculated with the IPCC 2021 method for climate change. The calculations were carried out in Brightway v2.4.6⁷¹. The assessment aims to evaluate the cradle-to-grave life cycle impacts of obtaining one kilogram of end product portfolio (i.e., mix of light gases, gasoline, diesel, and motor-oil). For the chemical recycling scenario, the LCI considers: (i) the complete life cycle of PP, including the extraction of raw materials, the PP production steps, the use phase of PP, and final waste collection and sorting; and (ii) the extraction of raw materials and production of materials/energy vectors needed in the hydrogen-free chemical recycling. The business-as-usual, BAU scenario consists of an equivalent multi-product system which accounts for the fossil-based production of the respective hydrocarbon products, as well as the production and end-of-life treatment of the corresponding PP (assuming that 66% of the waste polymer is incinerated and 34% landfilled)⁷². The use of fuels is excluded from the assessment, as combustion-related emissions are identical in both scenarios. Supplementary Tables 17–20 contain the LCI data used in the LCA calculations. The contributions of processes along the chemical recycling supply chain for shampoo lids and BAU pathways are shown in Supplementary Figs 36, 37. The cradle-to-grave system boundaries

considered for the circular waste plastic recycling and fossil-based BAU alternatives are displayed in Supplementary Fig. 39.

Additionally, an uncertainty assessment was performed for the chemical recycling and BAU scenarios. To this end, we applied Monte Carlo sampling with 500 iterations, based on the Ecoinvent pedigree matrix, which provides uncertainty levels for the life cycle inventory parameters. Uncertainties affecting economic performance were modelled using minimum and maximum values from the literature, as detailed in Supplementary Tables 17, 18. Results for all waste PP sources, including uncertainties in cost and GHG emissions, are shown in Supplementary Fig. 38.

Comparison across polyolefin chemical recycling technologies

The technoeconomic and environmental framework was adapted to assess alternative chemical recycling processes. A process simulation capable of handling different experimental datasets was carried out for the hydrotreatment technologies applied to waste plastic valorization (Supplementary Fig. 40). The mass yields used were taken from representative reports (Supplementary Table 22). This analysis was performed to identify the environmental and economic outlook of recycling waste plastics using the hydrogen-free depolymerization of polypropylene over $W_{10}Zr_{90}$ and other catalytic approaches, thereby helping to determine the best strategies for different plastic waste streams.

The assessment follows the same standardized procedure described for the hydrogen-free pathway. Conversion data from each experiment were used in the process simulations to determine the mass and energy requirements for the TEA and LCA. The operating conditions from the experimental work (e.g., temperature, pressure, catalyst-to-plastic ratio) were considered for each simulation. The catalysts were modelled according to the descriptions provided in the original studies, as summarized in Supplementary Table 23. Due to scarcity of performance stability studies, a uniform lifetime of 1000 h of operation was assumed for all technologies and scenarios.

The TAC and GWP values were calculated following the established TEA and LCA methodology. The profit margin was then determined as the difference between the levelized TAC and the potential revenues from selling the products of the chemical recycling plant. The net GWP reduction is defined as the difference between the cradle-to-grave GWP of the chemical recycling route and its analogous fossil BAU system, corresponding to the conventional refinery-based production emissions (cradle-to-gate) of the same product portfolio plus the cradle-to-grave emissions from the production and end-of-life of the plastic.

Data availability

Additional datasets generated in this study are provided in the Supplementary Information and at the Zenodo repository [<https://zenodo.org/records/17840461>]⁷³. XAS data generated in this study have been deposited in the ESRF database under the Experiment session: A31-I-307 on beamline BM31 [<https://doi.org/10.1515/ESRF-DC-2420488713>]⁷⁴. Data are available from the corresponding authors upon request.

Code availability

This study did not generate new code.

References

- Bachmann, M. et al. Towards circular plastics within planetary boundaries. *Nat. Sustain.* **6**, 599–610 (2023).
- Ma, D. Transforming end-of-life plastics for a better world. *Nat. Sustain.* **6**, 1142–1143 (2023).
- Martín, A. J., Mondelli, C., Jaydev, S. D. & Pérez-Ramírez, J. Catalytic processing of plastic waste on the rise. *Chem* **7**, 1487–1533 (2021).
- Sun, J. et al. Catalytic upcycling of polyolefins. *Chem. Rev.* **124**, 9457–9579 (2024).
- Li, H. et al. Expanding plastics recycling technologies: chemical aspects, technology status and challenges. *Green. Chem.* **24**, 8899–9002 (2022).
- Cardona, S. C. & Corma, A. Tertiary recycling of polypropylene by catalytic cracking in a semibatch stirred reactor: use of spent equilibrium FCC commercial catalyst. *Appl. Catal. B Environ.* **25**, 151–162 (2000).
- Cardona, S. C. & Corma, A. Kinetic study of the catalytic cracking of polypropylene in a semibatch stirred reactor. *Catal. Today* **75**, 239–246 (2002).
- Dong, Z. et al. Efficient conversion of polyethylene to light olefins by self-confined cracking and reforming. *Nat. Commun.* **16**, 7964 (2025).
- Vollmer, I. et al. Beyond mechanical recycling: giving new life to plastic waste. *Angew. Chem. Int. Ed.* **59**, 15402–15423 (2020).
- Liu, S., Kots, P. A., Vance, B. C., Danielson, A. & Vlachos, D. G. Plastic waste to fuels by hydrocracking at mild conditions. *Sci. Adv.* **7**, eabf8283 (2025).
- Wang, M. et al. Complete hydrogenolysis of mixed plastic wastes. *Nat. Chem. Eng.* **1**, 376–384 (2024).
- Wang, X. et al. Enhancing waste plastic hydrogenolysis on Ru/CeO₂ through concurrent incorporation of Fe single atoms and FeO_x nanoclusters. *Angew. Chem. Int. Ed.* **64**, e202506035 (2025).
- Rorrer, J. E., Beckham, G. T. & Román-Leshkov, Y. Conversion of polyolefin waste to liquid alkanes with Ru-based catalysts under mild conditions. *JACS Au* **1**, 8–12 (2021).
- Selvam, E. et al. Conversion of compositionally diverse plastic waste over earth-abundant sulfides. *J. Am. Chem. Soc.* **147**, 11227–11238 (2025).
- Zhang, Z. et al. Mixed plastics wastes upcycling with high-stability single-atom Ru catalyst. *J. Am. Chem. Soc.* **145**, 22836–22844 (2023).
- Jaydev, S. D., Usteri, M. E., Martín, A. J. & Pérez-Ramírez, J. Identifying selective catalysts in polypropylene hydrogenolysis by decoupling scission pathways. *Chem. Catal.* **3**, 100564 (2023).
- Jaydev, S. D., Martín, A. J., García, D., Chikri, K. & Pérez-Ramírez, J. Assessment of transport phenomena in catalyst effectiveness for chemical polyolefin recycling. *Nat. Chem. Eng.* **1**, 565–575 (2024).
- Guironnet, D. & Peters, B. Tandem catalysts for polyethylene upcycling: a simple kinetic model. *J. Phys. Chem. A* **124**, 3935–3942 (2020).
- Zhang, W. et al. Low-temperature upcycling of polyolefins into liquid alkanes via tandem cracking-alkylation. *Science* **379**, 807–811 (2023).
- Zhang, F. et al. Polyethylene upcycling to long-chain alkylaromatics by tandem hydrogenolysis/aromatization. *Science* **370**, 437–441 (2020).
- Sun, J. et al. Bifunctional tandem catalytic upcycling of polyethylene to surfactant-range alkylaromatics. *Chem.* **9**, 2318–2336 (2023).
- Du, J. et al. Efficient solvent- and hydrogen-free upcycling of high-density polyethylene into separable cyclic hydrocarbons. *Nat. Nanotechnol.* **18**, 772–779 (2023).
- Cen, Z. et al. Upcycling of polyethylene to gasoline through a self-supplied hydrogen strategy in a layered self-pillared zeolite. *Nat. Chem.* **16**, 871–880 (2024).
- Han, W. et al. One-pot catalytic conversion of polyethylene wastes to gasoline through a dual-catalyst system. *Chem.* **11**, 102340 (2025).
- Jeswani, H. et al. Life cycle environmental impacts of chemical recycling via pyrolysis of mixed plastic waste in comparison with mechanical recycling and energy recovery. *Sci. Total Environ.* **769**, 144483 (2021).
- Tomatis, M. et al. Environmental assessment of a novel ionic-liquid based method for recycling of PVC in composite materials. *Sci. Total Environ.* **887**, 163999 (2023).

27. Hernández, B., Kots, P., Selvam, E., Vlachos, D. G. & Ierapetritou, M. G. Techno-economic and life cycle analyses of thermochemical upcycling technologies of low-density polyethylene waste. *ACS Sustain. Chem. Eng.* **11**, 7170–7181 (2023).
28. Klotz, M., Oberschelp, C., Salah, C., Subal, L. & Hellweg, S. The role of chemical and solvent-based recycling within a sustainable circular economy for plastics. *Sci. Total Environ.* **906**, 167586 (2024).
29. Ross-Medgaarden, E. I. et al. New insights into the nature of the acidic catalytic active sites present in ZrO₂-supported tungsten oxide catalysts. *J. Catal.* **256**, 108–125 (2008).
30. Zhou, W. et al. Identification of active Zr–WO_x clusters on a ZrO₂ support for solid acid catalysts. *Nat. Chem.* **1**, 722–728 (2009).
31. Zhou, W. et al. Nature of catalytically active sites in the supported WO₃/ZrO₂ solid acid system: a current perspective. *ACS Catal.* **7**, 2181–2198 (2017).
32. Kwon, T., Kim, J. & Ro, I. Hydrogen-free catalytic strategies for the upcycling of polyolefin plastics. *Korean J. Chem. Eng.* **43**, 109–146 (2026).
33. Smeyers, B. et al. Reliability of milligram-scale high-throughput screening for evaluating polyolefin chemical recycling in the melt with heterogeneous catalysis. *Cell Rep. Phys. Sci.* **6**, 102981 (2025).
34. Shaker, M. & Rabnawaz, M. Tailoring olefin contents formation during waste plastic pyrolysis. *ACS Appl. Polym. Mater.* **6**, 9866–9874 (2024).
35. Hergesell, A. H. et al. Mechano-catalytic conversion of polypropylene over zeolite-based materials. *Catal. Sci. Technol.* **15**, 7525–7538 (2025).
36. Stephan, D. W. Frustrated Lewis pairs. *J. Am. Chem. Soc.* **137**, 10018–10032 (2015).
37. Zhao, X. et al. Selective hydrogenolysis of glycerol to 1,3-propanediol: manipulating the frustrated Lewis pairs by introducing gold to Pt/WO_x. *ChemSusChem* **10**, 819–824 (2017).
38. Li, J., Li, G. & Tsang, S. C. E. Heterogeneous frustrated Lewis pair catalysts: rational structure design and mechanistic elucidation based on intrinsic properties of supports. *Acc. Chem. Res.* **58**, 555–569 (2025).
39. Pinheiro Araújo, T. et al. Low-nuclearity CuZn ensembles on ZnZrO_x catalyze methanol synthesis from CO₂. *Nat. Commun.* **15**, 3101 (2024).
40. Pinheiro Araújo, T. et al. Flame spray pyrolysis as a synthesis platform to assess metal promotion in In₂O₃-catalyzed CO₂ hydrogenation. *Adv. Energy Mater.* **12**, 2103707 (2022).
41. Pinheiro Araújo, T. et al. Design of flame-made ZnZrO_x catalysts for sustainable methanol synthesis from CO₂. *Adv. Energy Mater.* **13**, 2204122 (2023).
42. Soultanidis, N. et al. Relating *n*-pentane isomerization activity to the tungsten surface density of WO_x/ZrO₂. *J. Am. Chem. Soc.* **132**, 13462–13471 (2010).
43. Tan, J. Z., Hullfish, C. W., Zheng, Y., Koel, B. E. & Sarazen, M. L. Conversion of polyethylene waste to short chain hydrocarbons under mild temperature and hydrogen pressure with metal-free and metal-loaded MFI zeolites. *Appl. Catal. B Environ.* **338**, 123028 (2023).
44. Jaydev, S. D. et al. Consumer grade polyethylene recycling via hydrogenolysis on ultrafine supported ruthenium nanoparticles. *Angew. Chem. Int. Ed.* **63**, e202317526 (2024).
45. Vance, B. C. et al. Single pot catalyst strategy to branched products via adhesive isomerization and hydrocracking of polyethylene over platinum tungstated zirconia. *Appl. Catal. B Environ.* **299**, 120483 (2021).
46. Monclús, L. et al. Mapping the chemical complexity of plastics. *Nature* **643**, 349–355 (2025).
47. Horsley, J. A., Wachs, I. E., Brown, J. M., Via, G. H. & Hardcastle, F. D. Structure of surface tungsten oxide species in the tungsten trioxide/alumina supported oxide system from X-ray absorption near-edge spectroscopy and Raman spectroscopy. *J. Phys. Chem.* **91**, 4014–4020 (1987).
48. Yamazoe, S., Hitomi, Y., Shishido, T. & Tanaka, T. XAFS study of tungsten L1- and L3-edges: structural analysis of WO₃ species loaded on TiO₂ as a catalyst for photo-oxidation of NH₃. *J. Phys. Chem. C.* **112**, 6869–6879 (2008).
49. Wachs, I. E., Kim, T. & Ross, E. I. Catalysis science of the solid acidity of model supported tungsten oxide catalysts. *Catal. Today* **116**, 162–168 (2006).
50. Ross-Medgaarden, E. I. & Wachs, I. E. Structural determination of bulk and surface tungsten oxides with UV–vis diffuse reflectance spectroscopy and Raman spectroscopy. *J. Phys. Chem. C.* **111**, 15089–15099 (2007).
51. Zou, S. et al. ‘Tearing effect’ of alloy–support interaction for alloy redispersion in NiRu/TiO₂ hydrogenation catalysts. *Angew. Chem. Int. Ed.* **64**, e202425066 (2025).
52. Ghosh, P., Priolkar, K. R. & Patra, A. Understanding the local structures of Eu and Zr in Eu₂O₃ doped and coated ZrO₂ nanocrystals by EXAFS study. *J. Phys. Chem. C.* **111**, 571–578 (2007).
53. Wang, C. et al. Polyethylene hydrogenolysis at mild conditions over ruthenium on tungstated zirconia. *JACS Au.* **1**, 1422–1434 (2021).
54. Salusso, D., Borfecchia, E. & Bordiga, S. Combining X-ray diffraction and X-ray absorption spectroscopy to unveil Zn local environment in Zn-Doped ZrO₂ catalysts. *J. Phys. Chem. C.* **125**, 22249–22261 (2021).
55. Ge, Y. et al. Defective zirconia promotes monometallic iron catalysts for higher alcohol synthesis. *Chem. Catal.* **4**, 101010 (2024).
56. Vollmer, I. et al. Unravelling potential reaction intermediates during catalytic pyrolysis of polypropylene with microscopy and spectroscopy. *Catal. Sci. Technol.* **14**, 894–902 (2024).
57. Hu, Q. et al. Polyethylene hydrogenolysis by dilute RuPt alloy to achieve H₂-pressure-independent low methane selectivity. *Nat. Commun.* **15**, 10573 (2024).
58. Mamede, A.-S. et al. Characterization of WO_x/CeO₂ catalysts and their reactivity in the isomerization of hexane. *J. Catal.* **223**, 1–12 (2004).
59. Wu, H. et al. Atomically engineered interfaces inducing bridging oxygen-mediated deprotonation for enhanced oxygen evolution in acidic conditions. *Nat. Commun.* **15**, 10315 (2024).
60. Whittingham, M. S. Hydrogen motion in oxides: from insulators to bronzes. *Solid State Ion.* **168**, 255–263 (2004).
61. Lalik, E. et al. Hydrogen spillover in tungsten oxide bronzes as observed by broadband neutron spectroscopy. *Energies* **16**, 5496 (2023).
62. Nogueroles-Langa, I. et al. Polyethylene hydrogenolysis to liquid products over bimetallic catalysts with favorable environmental footprint and economics. *Nat. Commun.* **16**, 9791 (2025).
63. Araújo, T. P. et al. Reaction-induced metal-metal oxide interactions in Pd-In₂O₃/ZrO₂ catalysts drive selective and stable CO₂ hydrogenation to methanol. *Angew. Chem. Int. Ed.* **62**, e202306563 (2023).
64. van Beek, W., Olga, S., Geir, W. & Emerich, H. SNBL, a dedicated beamline for combined in situ X-ray diffraction, X-ray absorption and Raman scattering experiments. *Ph. Transit.* **84**, 726–732 (2011).
65. Salah, C., Cobo, S., Pérez-Ramírez, J. & Guillén-Gosálbez, G. Environmental sustainability assessment of hydrogen from waste polymers. *ACS Sustain. Chem. Eng.* **11**, 3238–3247 (2023).
66. Ma, J. et al. Economic evaluation of infrastructures for thermochemical upcycling of post-consumer plastic waste. *Green. Chem.* **25**, 1032–1044 (2023).
67. Towler, G. & Sinnott, R. K. *Chemical Engineering Design* (Butterworth-Heinemann, 2021).
68. International Standards Organization. *Environmental management - Life cycle assessment - Principles and framework*. ISO 14040 (International Standards Organization, 2006).

69. International Standards Organization. *Environmental management - Life cycle assessment - Principles and framework. ISO 14044* (International Standards Organization, 2006).
70. Wernet, G. et al. The ecoinvent database version 3 (part I): overview and methodology. *Int. J. Life Cycle Assess.* **21**, 1218–1230 (2016).
71. Mutel, C. Brightway: an open source framework for life cycle assessment. *J. Open Source Softw.* **2**, 236 (2017).
72. Ragaert, K., Delva, L. & Van Geem, K. Mechanical and chemical recycling of solid plastic waste. *Waste Manag.* **69**, 24–58 (2017).
73. Zou, S. et al. Interfacial W–O–Zr ensembles in tungstated zirconia catalysts enable efficient hydrogen-free recycling of polypropylene waste, Zenodo, <https://zenodo.org/records/17840461> (2026).
74. Ge, Y., Zou, S., Martín, A. J., & Pérez-Ramírez, J. (2026) Operando XAS investigation of W–O–Zr ensembles in tungstated zirconia for hydrogen-free recycling of polypropylene waste (Version 1) [Dataset]. European Synchrotron Radiation Facility. <https://doi.org/10.1515/ESRF-DC-2420488713>.
75. Keramidas, V. G. & White, W. B. Raman scattering study of the crystallization and phase transformations of ZrO₂. *J. Am. Ceram. Soc.* **57**, 22–24 (1974).
76. Thackeray, D. P. C. The Raman spectrum of zirconium dioxide. *Spectrochim. Acta Part A Mol. Spectrosc.* **30**, 549–550 (1974).

Acknowledgements

The authors are grateful to Yung-Tai Chiang for Raman measurements, Manuel Meinhardt for contributing to catalyst evaluation, and Dr. Thayan Pinheiro Araújo for initial work on the reactor setup and preliminary tests. Electron Microscopy was performed at the Scientific Center for Optical and Electron Microscopy (ScopeM, ETH Zürich). The Swiss-Norwegian beamlines (SNBL, ESRF) are acknowledged for provision of beamtime and its staff for invaluable support.

Author contributions

S.Z. conducted most of the catalytic evaluations, characterizations, operando experiments, and data analysis. Y.G. synthesized WZr/KIT-6 + HZSM-5 catalyst, performed additional catalytic evaluations and data analysis, and designed and supervised the operando experiments. P.O.W. synthesized the tungstated zirconia catalysts by flame spray pyrolysis under the supervision of R.N.G. and W.J.S. J.F.-G. conducted environmental and technoeconomic assessments under the supervision of G.G.-G. A.J.M. coordinated and supervised the study. J.P.-R. conceptualized and supervised the entire project and managed resources and funding. S.Z. and A.J.M. drafted the manuscript, S.Z., Y.G., A.J.M., and J.P.-R. revised the manuscript. All authors edited and approved the final version of the manuscript.

Funding

This publication was created as part of NCCR Catalysis (grant number 225147), a National Centre of Competence in Research funded by the Swiss National Science Foundation (J.P.-R.).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-73420-6>.

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Peer review information *Nature Communications* thanks Mohamed Shaker and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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